Huawei MG-ACC-46 Access Node QoS Congestion Event Resolution SOP

Node: MG-ACC-46 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_1899

Description

The occurrence of the QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) failure domain on the MG-ACC-46 access node at the MG Road Core site is primarily attributed to the inability of the Huawei access node's QoS policy to effectively manage and prioritize traffic, resulting in congestion on the interface. This is further exacerbated by the limitations of the Huawei access node's queue management and scheduling algorithms. The underlying protocol mechanics, specifically the RSVP-TE (Resource Reservation Protocol - Traffic Engineering) protocol, also play a significant role in this failure domain. The RSVP-TE protocol is used to establish and manage LSPs (Label Switched Paths) on the MG-ACC-46 node, and any issues with the RSVP-TE protocol can lead to congestion on the interface. Furthermore, the hardware mechanics of the Huawei access node, specifically the TCAM (Ternary Content Addressable Memory) and the FPC (Flexible PIC Concentrator), can also contribute to this failure domain. The TCAM is used to store and manage the forwarding information base (FIB) on the MG-ACC-46 node, and the FPC is used to concentrate the traffic on the interface. Any issues with the TCAM or the FPC can lead to congestion on the interface.

Pre-requisite Checks

- check: Dispatch remote hands to MG Road Core for physical access to the MG-ACC-46 node., description: Ensure that remote hands are dispatched to MG Road Core for physical access to the MG-ACC-46 node before proceeding with the diagnostic and resolution steps.

- check: Verify the status of the redundant link on the MG-ACC-46 node., description: Verify that the redundant link on the MG-ACC-46 node is functional and not causing any congestion on the interface.

Potential Root Causes

Cause	Probability
QoS policy misconfiguration on the MG-ACC-46 node.	High
Limitations of the Huawei access node's queue management and scheduling algorithms.	Medium
Issues with the RSVP-TE protocol on the MG-ACC-46 node.	Medium
Hardware issues with the TCAM or the FPC on the MG-ACC-46 node.	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos police, Expected Output: Display the QoS police configuration on the MG-ACC-46 node., Description: This command displays the QoS police configuration on the MG-ACC-46 node, which includes the policing rates and burst sizes. This information can be used to determine if the QoS policy is misconfigured.
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Command: display qos queue, Expected Output: Display the QoS queue configuration on the MG-ACC-46 node., Description: This command displays the QoS queue configuration on the MG-ACC-46 node, which includes the queue lengths and scheduling algorithms. This information can be used to determine if the Huawei access node's queue management and scheduling algorithms are functioning correctly.

Command: display rsvp-te database, Expected Output: Display the RSVP-TE database on the MG-ACC-46 node., Description: This command displays the RSVP-TE database on the MG-ACC-46 node, which includes the LSPs and their associated attributes. This information can be used to determine if there are any issues with the RSVP-TE protocol.

Command: display tcam, Expected Output: Display the TCAM configuration on the MG-ACC-46 node., Description: This command displays the TCAM configuration on the MG-ACC-46 node, which includes the TCAM size and the number of entries. This information can be used to determine if there are any issues with the TCAM.

Command: display fpc, Expected Output: Display the FPC configuration on the MG-ACC-46 node., Description: This command displays the FPC configuration on the MG-ACC-46 node, which includes the FPC status and the number of interfaces. This information can be used to determine if there are any issues with the FPC.
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Resolution Steps

Step 1: Step: Reconfigure the QoS policy on the MG-ACC-46 node., Command: undo qos police, Description: Reconfigure the QoS policy on the MG-ACC-46 node by removing any misconfigured policing rates or burst sizes. This can be done using the `undo qos police` command.

Step 2: Step: Verify the QoS queue configuration on the MG-ACC-46 node., Command: display qos queue, Description: Verify the QoS queue configuration on the MG-ACC-46 node by checking the queue lengths and scheduling algorithms. This can be done using the `display qos queue` command.

Step 3: Step: Reconfigure the RSVP-TE protocol on the MG-ACC-46 node., Command: undo rsvp-te, Description: Reconfigure the RSVP-TE protocol on the MG-ACC-46 node by removing any misconfigured LSPs or attributes. This can be done using the `undo rsvp-te` command.

Step 4: Step: Verify the TCAM configuration on the MG-ACC-46 node., Command: display tcam, Description: Verify the TCAM configuration on the MG-ACC-46 node by checking the TCAM size and the number of entries. This can be done using the `display tcam` command.

Step 5: Step: Verify the FPC configuration on the MG-ACC-46 node., Command: display fpc, Description: Verify the FPC configuration on the MG-ACC-46 node by checking the FPC status and the number of interfaces. This can be done using the `display fpc` command.

Post-Check Verification

```
Command: display qos police, Expected Output: Verify that the QoS police configuration on the MG-ACC-46 node is correctly configured., Description: Verify that the QoS police configuration on the MG-ACC-46 node is correctly configured by checking the policing rates and burst sizes. This can be done using the `display qos police` command.

Command: display qos queue, Expected Output: Verify that the QoS queue configuration on the MG-ACC-46 node is correctly configured., Description: Verify that the QoS queue configuration on the MG-ACC-46 node is correctly configured by checking the queue lengths and scheduling algorithms. This can be done using the `display qos queue` command.

Command: display rsvp-te database, Expected Output: Verify that the RSVP-TE database on the MG-ACC-46 node is correctly configured., Description: Verify that the RSVP-TE database on the MG-ACC-46 node is
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correctly configured by checking the LSPs and their associated attributes. This can be done using the `display rsvp-te database` command.

Command: display tcam, Expected Output: Verify that the TCAM configuration on the MG-ACC-46 node is correctly configured., Description: Verify that the TCAM configuration on the MG-ACC-46 node is correctly configured by checking the TCAM size and the number of entries. This can be done using the `display tcam` command.

Command: display fpc, Expected Output: Verify that the FPC configuration on the MG-ACC-46 node is correctly configured., Description: Verify that the FPC configuration on the MG-ACC-46 node is correctly configured by checking the FPC status and the number of interfaces. This can be done using the `display fpc` command.
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Roll-back Procedures

- Step: Revert the QoS policy configuration on the MG-ACC-46 node., Command: undo qos police, Description: Revert the QoS policy configuration on the MG-ACC-46 node to the previous state by removing the changes made in the resolution steps.

- Step: Revert the RSVP-TE protocol configuration on the MG-ACC-46 node., Command: undo rsvp-te, Description: Revert the RSVP-TE protocol configuration on the MG-ACC-46 node to the previous state by removing the changes made in the resolution steps.

- Step: Revert the TCAM configuration on the MG-ACC-46 node., Command: undo tcam, Description: Revert the TCAM configuration on the MG-ACC-46 node to the previous state by removing the changes made in the resolution steps.

- Step: Revert the FPC configuration on the MG-ACC-46 node., Command: undo fpc, Description: Revert the FPC configuration on the MG-ACC-46 node to the previous state by removing the changes made in the resolution steps.